

Batteryless E-PAPER DISPLAY

ASHWINI KUMAR SINHA

As technology advances, various energy harvesting methods have emerged, leading to a new era of batteryless sensors. The batteryless E-paper display proposed here is one such design. (Electronic paper, also known as electronic ink (e-ink) or intelligent paper, is a display device that mimics the appearance of ordinary ink on paper.) It might be surprising, but it is no longer considered rocket science or science fiction. Fig. 1 showcases the author's prototype of a batteryless e-ink display.

Many of you may have already seen or used batteryless e-ink displays (E-paper displays). Several companies are already selling them, a few of which can be checked here:

1. https://www.waveshare.com/wiki/7.5inch_NFC-Powered_e-Paper
2. <https://www.ynvisible.com/news-inspiration/ynvisible-e-paper-displays-nfc-energy-harvesting>
3. <https://www.good-display.com/product/423.html>
4. <https://lummico.com/product/smarttag-batteryless-2-90>
5. <https://www.usocard.com/case-detail/id-32.html>
6. <https://github.com/GDSID/Hipointk>

e-ink display

e-ink displays are a unique kind of display where once you update the screen, the displayed content remains even if you remove the power. Power is only needed when updating, and the required amount is minimal, which can be easily supplied by harvesting RF, solar, and various other forms of energy in the vicinity.



Fig. 1: Batteryless e-ink display prototype

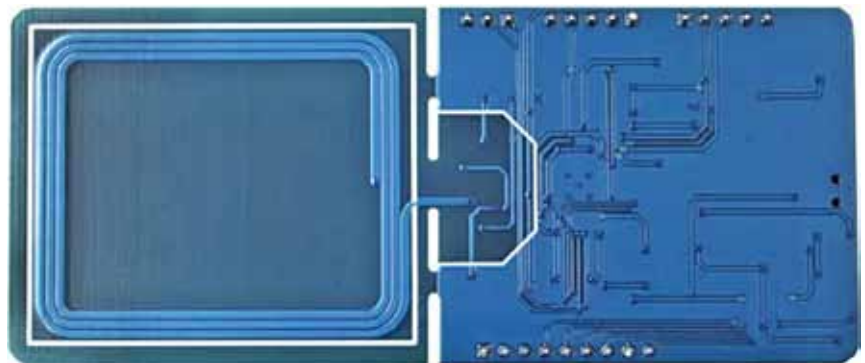


Fig. 2: Energy harvesting display driver module

TABLE 1
BILL OF MATERIALS FOR NFC-BASED BATTERYLESS DISPLAY

Component	Description	Quantity
DENFC-M0 energy harvesting board	SPI module DENFC-M01	1
GDEY0213B74 e-ink display	1.5s, partial refresh, 4 Grayscale, IC SSD1680Z, resolution 250x122	1
ST link emulator	Connector module	1